`forestgap` 1.1 Experiment Protocol

Protocol version: `forestgap-experiment-program/1.1` **Status:** normative for new benchmark results; legacy exploratory notebooks and pre-1.1 result files are not evidence for this protocol.

This document defines the executable study program for `forestgap` 1.1. A result is in scope only when its declared estimator, relation, data split, tuning trial, preprocessing, and status are captured in a manifest row. The shared harness is the source of truth for those rows. A failed, unsupported, or excluded requested comparison is retained as data; it is never silently dropped.

Scope and capability matrix

Cells are generated from `capabilities_for(fitted_estimator)`, not from a legacy hard-coded relation list. The static registry remains the roadmap, while the fitted capability snapshot is the executable contract. Do not schedule gated or unregistered families. Survival is explicitly deferred in this phase.

Family	Task scope	Relations eligible after the runtime contract	Study status
RF, ExtraTrees, RotationForest, treeple oblique/patch/extra-oblique, CCF/CCF-Bag	classification, regression	`original`, `oob`, `rfgap`, `kerf`, `path`, `englund`, `random_height`	executable when installed and bootstrap/OOB metadata satisfy the relation contract
HonestForest classifier	classification	`original`, `honest_gap`, `kerf`, `path`, `englund`, `random_height`	executable when honesty-role indices are available
GradientBoosting, XGBoost, LightGBM, CatBoost	classification, regression	`original`, `kerf`, `boosted`, `path`, `englund`, `random_height`	executable; CatBoost path-derived relations require `grow_policy="SymmetricTree"`
Quantile forest	regression	`conditional`, `original`, `kerf`, `path`, `englund`, `random_height`	executable when the optional adapter is installed
Isolation and extended-isolation forest	anomaly	`collision`	executable; native path scores remain separate comparators

Family	Task scope	Relations eligible after the runtime contract	Study status
Synthetic-null forest	anomaly, geometry	`original`, `oob`, `kerf`, `path`, `englund`, `random_height`	executable; RF-GAP is intentionally absent
Random survival forest	survival	`survival_leaf`	deferred: no benchmark runner, dataset dependency, or result claim
RF-AE / autoencoding, causal, generalized, Bayesian, structured, multiview, graph, and other gated families	varies	none	excluded until their adapter and data-specific protocol exist

`oob` and `rfgap` require the fitted bootstrap, in-bag, and OOB contracts. Boosting settings never create those contracts. Cells requested for a relation outside a fitted snapshot are recorded as `unsupported`; deliberately deferred, gated, or unregistered cells are recorded as `excluded` without fitting the factory.

Directionality

`rfgap`, `honest_gap`, and `conditional` are directed query-to-training weights. Use them directly for prediction, conditional distribution/quantile work, imputation donors, retrieval, active acquisition, and attribution. Symmetrize a directed relation only in a declared geometry application (embedding, clustering, or graph diagnostic), and record that policy in the manifest. `original`, `oob`, `kerf`, `boosted`, `path`, `englund`, `random_height`, and `collision` are symmetric relations under their documented contracts. Prediction identity is evaluated only for RF-GAP and Honest-GAP.

Application × model × relation declaration

The family matrix above says what a fitted model can expose. This table says which of those fitted capabilities each proximity-based application requests. “All fitted relations” means exactly the runtime `capabilities_for` snapshot for the listed family; it does not mean a legacy, hard-coded relation list. Thus unavailable optional backends and missing bootstrap/OOB/honesty metadata produce retained `unsupported` rows rather than a substituted relation.

Application / runner	Models included	Relations requested and consumption rule
Kernel/kNN prediction and retrieval (`run_protocol_matrix.py`,	each executable family exposed by the	all fitted relations; `rfgap`, `honest_gap`, and

Application / runner	Models included	Relations requested and consumption rule
`run_method_matrix.py`)	capability-aware runner (bagged, HonestForest, boosted, and quantile forest as installed)	`conditional` are used directly as query-to-training weights, all other relations as their symmetric weights
Prediction identity (`run_prediction_identity.py`)	RF-GAP-capable bagged families and HonestForest	`rfgap` for bagged families; `honest_gap` for HonestForest; no other relation is an identity comparison
Quantiles and conformal	quantile forest only	`conditional` only, plus native quantile predictions and split conformal comparators; ordinary co-leaf relations are not claimed as conditional-distribution weights
Imputation (`run_imputation.py`)	executable bagged and HonestForest families in the legacy runner; all executable families in capability-aware studies where donor prediction is supported	all fitted relations for one-pass and iterative donors; directed relations remain directed donor weights
Local feature importance and feature selection	each supervised family supported by the runner	all fitted relations; perturbation recomputes the same selected relation against the fixed training reference
Prototypes and retrieval (`run_prototypes.py`)	each supervised classification family supported by the runner	all fitted relations; any directed matrix is symmetrized explicitly for center-to-center geometry, while query retrieval stays directed
Visualization, embedding, and clustering (`run_mds_visualization.py`, `run_method_matrix.py`)	each executable family supported by the runner; the capability-aware matrix is the cross-family coverage runner	all fitted relations; `rfgap`, `honest_gap`, and `conditional` are symmetrized with the recorded geometry policy before affinity/distance conversion

Application / runner	Models included	Relations requested and consumption rule
Anomaly detection (`run_outlier_detection.py`)	isolation/extended-isolation forests and synthetic-null forests; supervised forests only as declared comparators	isolation families: `collision`; synthetic-null: its fitted symmetric relations; supervised comparators: `original`, `oob`, `rfgap`, `kerf`, `path`, `englund`, and `random_height`. RF-GAP is included; Honest-GAP remains excluded.
Active learning	executable supervised predictive families	all fitted relations, with directed relations used directly for predictive/acquisition weights
Training-instance attribution	RF-GAP-capable bagged families, HonestForest, and quantile forest	`rfgap`, `honest_gap`, and `conditional`, respectively; no symmetric relation is treated as a directed attribution weight
Alignment	RF-GAP-capable bagged families only	`rfgap`; symmetrization is prohibited for label transfer/attribution and separately recorded if geometry diagnostics require it

The capability-aware runners enforce these scopes through `experiments.program.APPLICATION_RELATION_SCOPES`. Legacy runners retain their model-specific candidate rows so that unavailable optional models and invalid runtime contracts remain visible in their outputs.

Dataset and split manifests

The package exposes typed `OpenMLTask` and `OpenMLTaskFold` objects. These hold the task target, repeat/fold/sample dimensions, official train/test indices, and a stable `r{repeat}-f{fold}-s{sample}` fold identifier.

Study family	Dataset source	Outer evaluation protocol
Supervised, imputation, explanations, selection,	OpenML-CC18 classification study 99	every eligible task and every official OpenML repeat/fold/sample; never

Study family	Dataset source	Outer evaluation protocol
prototypes, retrieval, active learning, attribution	and OpenML study 455 regression	regenerate an outer `train_test_split`
Geometry and clustering	eligible ClustBench v1.1 datasets plus controlled simulations	label-free fitting/tuning; labels are outer evaluation only
Anomaly	all 57 eligible ADBench datasets plus controlled simulations	labels are outer scoring only for label-free runs
RF-MA and FoSTA reference tracks	their documented UCI view-split datasets	preserve the reference study's views, masks, repetitions, and anchor rules
RF-MA-style regression extension	paired feature views created only from study-455 outer-training data	30% training-only anchors; target labels hidden from fitting and tuning

The prior regression suite, OpenML study 353 / CTR23, is not part of this protocol. Existing generic `load_openml_suite` support is retained for backwards compatibility; study runners use typed task/fold access instead.

Every cell records at least: protocol version; suite, task, dataset, repeat, fold, sample, and fold IDs; split and artifact hashes; parent and relation trials; preprocessing/calibration roles; runtime capability snapshot; relation settings and directionality policy; configuration; metrics; dependency versions; code revision; fit/relation/application timing; peak memory; and a terminal `ok`, `unsupported`, `excluded`, or `error` status with a reason when not `ok`.

Study portfolio

Track	Required comparisons	Primary outcomes
General prediction benchmarking	native parent, proximity kNN/kernel prediction, and identity where valid	balanced accuracy and log loss for classification; RMSE, MAE, and R^2 for regression
Quantiles and conformal	quantile forest/native quantiles, conditional proximity weights, split conformal	pinball loss, coverage, interval or set size
Imputation	median/mode, kNN, iterative imputation, one-pass and iterative proximity donors at MCAR/MAR/MNAR 5/10/25/50/75%	normalized RMSE/MAE, categorical accuracy, downstream retention

Track	Required comparisons	Primary outcomes
Explanations and selection	local importance versus permutation, SHAP, and LIME; feature selection versus mutual information and embedded baselines	deletion/insertion fidelity, synthetic truth recovery, stability
Prototypes and retrieval	learned prototypes and non-proximity comparators	prototype coverage/purity/stability; Recall@k, mAP, nDCG, latency
Geometry and clustering	RF-PHATE/Isomap/UMAP-style embeddings; spectral, HDBSCAN, and kernel-PCA clustering	ARI/NMI/AMI, stability, trustworthiness, continuity, neighborhood preservation
Anomaly detection	native isolation scores, collision relation, synthetic-null forests, proximity outlier score, LOF, and kNN distance	ROC-AUC, PR-AUC/AP, Precision@contamination, runtime, stability
Active learning and attribution	random, uncertainty, core-set, and proximity-diverse acquisition; directed predictive attribution only	learning curves from 10% to 50% labels in 5% batches; deletion/retraining fidelity
Alignment	faithful RF-MA and FoSTA classification references; a separately named RF-MA-style regression extension	label transfer, FOSCTTM, alignment score, RMSE/MAE/R^2 where applicable, runtime, memory, and separate transductive/OOS reporting

Cluster count, embedding hyperparameters, and all geometry tuning use only training covariates, graph/reconstruction diagnostics, and stability. Labels may colour plots or score the held-out result, but may not choose a label-free configuration. The method-matrix smoke runner uses a data-only eigengap criterion and records the selected count.

Tuning and refitting

For every application × outer fold:

1. Freeze a seeded 50-trial family-specific parent manifest and a seeded 25-trial relation/application manifest before fitting. The provided helpers use Sobol or Latin-hypercube draws and persist the exact values.
2. Use five inner folds that are subsets of the outer-training indices. Select the parent independently for each application.
3. Refit that selected parent once on the application outer-training partition and reuse that refit across eligible relation comparisons only.
4. Fit calibration roles, encoders, scalers, feature selection, missingness generators, view construction, and anchors inside the outer-training partition. The CatBoost-native

categorical branch is declared separately; it must not receive generic ordinal preprocessing.

The required conditional spaces are:

Family or component	Search space
Bagged forests	256–1,024 trees; depth {None, 8, 16, 32}; leaf size {1, 2, 5, 10, 20}; valid feature/sample fractions; enforce bootstrap for OOB/RF-GAP candidates
Boosters	100–1,000 stages; learning rate 0.01–0.3; depth/leaves; minimum child/leaf size; row/column sampling; CatBoost symmetric trees for path studies
Relation/application settings	Englund weight {0.25, 0.5, 1, 2, 4}; neighbours {3, 5, 10, 20, 40}; kernel regularization; imputation iterations {1, 3, 5, 10}; label-free geometry grids

Leakage controls

- Official outer test rows are never used in preprocessing, fitting, tuning, missingness generation, calibration, anchor selection, or model selection.
- Inner train/validation partitions and calibration partitions are disjoint and are expressed as original outer-training row IDs in the manifest.
- Missingness is induced only after the outer split; its mechanism and seed are recorded.
- Label-free tracks never inspect labels until outer evaluation. Label-informed geometry is a separately labelled analysis, never mixed with label-free results.
- Directed predictive relations are not silently symmetrized. Geometry symmetrization is explicit and recorded.
- Paired-view anchors are sampled only from training rows. FoSTA remains UCI-only; the regression extension is explicitly a new RF-MA-style study, not a FoSTA reproduction.

Analysis and acceptance criteria

Aggregate outer folds within each dataset before cross-dataset comparisons; do not pool raw fold scores. Report paired effects, bootstrap confidence intervals, average ranks, an omnibus test, and Holm-adjusted paired comparisons. Report runtime and peak memory alongside predictive or structural metrics.

A study is publishable under this protocol only when it has:

- one complete manifest row per requested cell, including clean optional-dependency skips;
- deterministic smoke coverage for every installed family/relation/application contract;
- synthetic checks for RF-GAP/Honest-GAP identity, imputation recovery, feature-truth recovery, cluster/anomaly recovery, and alignment label transfer;
- tests for official-fold loading, runtime capability generation, status preservation, manifest completeness, inner/outer isolation, calibration, directionality, label-free tuning, and training-only anchors; and

- an explicit statement of any deferred survival, RF-AE, single-cell, gated, or unavailable optional-dependency work.

This protocol incorporates the prior request for general prediction benchmarking while replacing the previous exploratory outline.